COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 - Data Structures and Algorithms	**SECTION:**	CPE21S1
DATE PERFORMED:	Oct 15, 2026	**DATE SUBMITTED:**	Oct 15, 2026
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ACTIVITY NO. 13.1 **[Parallel Algorithms and Multithreading]**

1. PROCEDURE OUTPUT

ILO A	

Screenshot	**Code:**
	`#include <iostream>`
	`#include <thread>`
	
	`void print(int n, const std::string &str){`
	`  std::cout << "Printing integer: "<< n <<std::endl;`
	`  std::cout<<" Printing string: "<< str << std::endl;`
	`}`
	
	`int main(){`
	
	`  std::thread t1(print,10,"T.I.P.");`
	`  t1.join();`
	
	`  return 0;`
	`}`
	
	Output:
	`Printing integer: 10`
	`Printing string: T.I.P.`

Analysis	Using the **<thread>** library it makes it so the program can run multithreading task. By executing the **print()** function within a worker thread **t1** , the main thread passes the value **10** and **T.I.P** to **t1**and implements **t1.join()** to wait until the worker finishes printing

Table 13-1. Simple One-Threaded Example

ILOB	

Screenshot of the code/Output	ILOB > Multithread.cpp > main()
	1 #include <iostream>
	2 #include <string>
	3 #include <vector>
	4 #include <thread>
	5
	6 void print(int n, const std::string &str) {
	7 std::string msg = std::to_string(n) + " : " + str;
	8 std::cout << msg << std::endl;
	9 }
	10
	11 int main() {
	12 std::vector<std::string> s = {
	13 "T.I.P.",
	14 "Competent",
	15 "Computer",
	16 "Engineers"
	17 };
	18
	19 std::vector<std::thread> threads;
	20
	21 for (int i = 0; i < s.size(); i++) {
	22 threads.push_back(std::thread(print, i, s[i]));
	23 }
	24
	25 for (auto &th : threads) {
	26 th.join();
	27 }
	28
	29 return 0;
	30 }

OUTPUT:	PS C:\Users\TIPQC\Desktop 0 : T.I.P.1 : Competent 2 : Computer 3 : Engineers PS C:\Users\TIPQC\Desktop\ 0 : T.I.P.2 : Computer 1 : Competent 3 : Engineers PS C:\Users\TIPQC\Desktop\ 1 : Competent0 : T.I.P. 3 : Engineers 2 : Computer PS C:\Users\TIPQC\Desktop\ PS C:\Users\TIPQC\Desktop\AL 3 : Engineers1 : Competent 2 : Computer 0 : T.I.P. 3 : Engineers0 : T.I.P. 1 : Competent 2 : Computer PS C:\Users\TIPQC\Desktop\A
Analysis	

Table 13-2. Multithreaded Example.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

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I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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